STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Type	Python-based data analysis (no machine learning)
Environment	Google Colab
Libraries	Python, NumPy, Pandas, Matplotlib
Dataset	500 synthetic student records

Abstract

The Student Performance Analytics System is a beginner-friendly Python project that generates and analyzes a dataset of 500 synthetic student records. It uses Pandas to organize tabular data, NumPy to generate numerical marks and support calculations, and Matplotlib to display selected results as charts. The system calculates totals and averages, assigns grades, determines pass/fail status using defined rules, ranks students, summarizes subject performance, and exports reports as CSV files. The project focuses on programming fundamentals and data analysis; it does not use machine learning.

1. Introduction

Reviewing many student records manually can be time-consuming. This project demonstrates how a Python program can organize student marks and attendance, calculate summary values, and present patterns in a consistent format. The dataset is generated for practice and does not represent verified real student information.

2. Objectives

- Generate 500 synthetic student records and save them in CSV format.
- Use Pandas to load, inspect, organize, and analyze the dataset.
- Use NumPy for numerical generation and calculations.
- Calculate total marks and average marks across five subjects.
- Assign grades and classify students as Pass or Fail using stated rules.
- Identify top performers and rank students by average marks.
- Compare average, highest, and lowest marks across subjects.
- Visualize selected results and export analysis reports to CSV.
- Practice variables, lists, dictionaries, loops, functions, and conditional statements.

3. Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language for dataset generation and analysis.
NumPy	Generates varied synthetic marks and supports numerical operations.
Pandas	Stores records in a DataFrame, inspects data, calculates summaries, and exports CSV files.
Matplotlib	Creates a subject-average bar chart, pass/fail pie chart, and average-mark histogram.
Google Colab	Browser-based environment used to write and run notebook cells.
CSV	Stores the dataset and exported analysis results in a portable tabular format.

4. Dataset Design

The exported notebook shows a dataset containing 500 rows and 9 original columns. Each row represents one synthetic student. Student IDs follow a format such as STU0001. The five subject marks are generated between 0 and 100, and attendance is represented as a percentage.

Column	Description
Student_ID	Unique synthetic student identifier.
Name	Synthetic full name.
Department	One of five sample academic departments.
Python, Mathematics, AI, Statistics, Communication	Marks for five subjects, each out of 100.
Attendance	Attendance percentage, generated from 50% to 100%.

5. Methodology

Dataset generation

A seeded NumPy random-number generator assigns performance groups and generates subject marks. Student names are selected from possible first-name/last-name combinations, and records are assembled using dictionaries and a for loop.

Data organization

The list of records is converted into a Pandas DataFrame and saved as students.csv.

Data quality checks

The notebook checks dataset dimensions, data types, missing values, duplicate student IDs, and whether marks and attendance fall within the allowed ranges.

Performance calculations

Total marks are calculated as the sum of the five subject marks. Average marks are calculated as the mean of those five marks.

Grade and result rules

Grades are based on average marks. A student passes only if every subject mark is at least 40; otherwise, the result is Fail.

Summary and ranking

The notebook counts pass/fail outcomes, summarizes grade distribution, identifies the highest and lowest average marks, lists the top 10 students, and compares subjects.

Visualization and export

Matplotlib charts summarize selected results, and Pandas exports the full performance report, subject report, and grade distribution report as CSV files.

6. Grading and Pass/Fail Rules

Average Marks	Grade
90–100	A+
80–89.99	A
70–79.99	B
60–69.99	C
50–59.99	D
Below 50	F

Pass rule: A student passes only when all five subject marks are 40 or above. One or more subject marks below 40 makes the overall result Fail. Grade and Pass/Fail are separate: a student can receive a low grade and still pass if all subject marks meet the threshold.

7. Results from the Exported Notebook

The values below are taken from the output visible in the provided notebook PDF. They describe the generated sample dataset, not real student performance.

Metric	Reported result
Student records	500
Original dataset dimensions	500 rows × 9 columns
Exported performance report dimensions	500 rows × 14 columns
Missing values	0
Duplicate student IDs	0
Pass count / percentage	375 / 75.0%
Fail count / percentage	125 / 25.0%
Highest average shown	94.4 (A+)
Lowest average shown	27.8 (F)

Grade distribution

Grade	Students
A+	16
A	94
B	71
C	165
D	38
F	116

Subject-wise average marks

Subject	Average	Highest	Lowest
Python	64.57	100	12
Mathematics	64.72	100	15
AI	64.73	100	19
Statistics	64.65	100	17
Communication	64.78	100	11

In the exported output, Communication has the highest class average (64.78), while Python has the lowest (64.57). The differences are small in this generated dataset.

8. Output Files

students.csv — The generated dataset containing 500 student records and 9 original fields.

student_performance_report.csv — The detailed report with derived calculations, grades, result, and rank (14 columns in the shown output).

subject_performance_report.csv — Average, highest, and lowest marks for each subject.

grade_distribution_report.csv — Count of students in each grade category.

Student_Performance_Analytics.ipynb — The Colab notebook containing code, explanations, and displayed outputs. Download the actual notebook from Colab for GitHub; the provided PDF is only an export.

9. How to Run the Project

1. Open the original notebook in Google Colab.
2. Select Runtime → Run all, or run cells from top to bottom.
3. Confirm that the dataset has 500 rows and that the quality checks pass.
4. Confirm that the charts are displayed and the three CSV reports are saved.
5. Download the notebook as an .ipynb file and download the generated CSV reports if they are to be includedin the repository.

10. Limitations

• All student data is synthetic; the results should not be interpreted as real educational findings.

• The project uses predefined grading and pass/fail rules. Different institutions may use different rules.

• The analysis is descriptive and rule-based. No machine-learning model is used.

11. Conclusion

The Student Performance Analytics System demonstrates a complete beginner-level data-analysis workflow: creating synthetic records, checking data quality, calculating marks and grades, ranking students, summarizing class and subject performance, visualizing results, and exporting CSV reports. It provides practice with Python fundamentals, NumPy, Pandas, and Matplotlib without introducing machine learning. The next improvement is to organize the notebook and exported files into a clean GitHub repository with a clear README and the actual .ipynb source.